Autonomous Trading Agent using Deep Reinforcement Learning

Andrew BEJJANI, Nael ARNOUX

DSBA, Reinforcement Learning Project

Introduction & Motivation

Financial market trading is a complex and dynamic task characterized by high volatility and non-stationary data. Traditional trading strategies often rely on rigid rules or linear models that struggle to adapt to rapidly changing market conditions. Recently, Deep Reinforcement Learning (DRL) has emerged as a promising approach for automated trading, allowing an agent to learn optimal policies directly from interactions with the market environment.

Project Objective: The ultimate goal is to build an autonomous system trading S&P 500 stocks (SPY) that generates profit with a superior risk-adjusted return compared to the market average, specifically targeting the maximization of the Sharpe Ratio.

Environment Modeling (MDP)

The trading problem is modeled as a Markov Decision Process (MDP), where the agent interacts with the market over time.

State Space ($\mathcal{S}$)

The state space uses historical market data alongside technical indicators to provide a comprehensive market view. Features include:

- Account Balance and Shares Held.
- SPY Close Price and Volatility Index (VIX).
- Technical Indicators: MACD, RSI (14-day), and Bollinger Bands (20-day).

To provide short-term memory, observations are stacked over 10 timesteps using `VecFrameStack`.

Action Space ($\mathcal{A}$)

The agent uses a continuous action space defined between 0.0 (100% cash) and 1.0 (100% stock). A hardcoded stop-loss is also triggered if a position loses more than 3% of its value.

Reward Function ($\mathcal{R}$)

To align with the objective of outperforming the market, the reward is based on the Logarithmic Alpha. It measures the difference between the portfolio's log return and the market's log return:

$$R_t = 1500 \times \left(\ln\left(\frac{NW_t}{NW_{t-1}}\right) - \ln\left(\frac{P_t}{P_{t-1}}\right) \right) \tag{1}$$

where NW is the Net Worth and P is the SPY close price. A fixed multiplier of 1500 is applied to stabilize network learning.

Agent Architectures & Training

We trained three distinct DRL algorithms over 500,000 timesteps each. The models were trained on 10 years of historical data (2015-2024) and tested on an unseen 1-year out-of-sample period (2025).

1. **A2C (Advantage Actor-Critic):** A synchronous, deterministic variant of A3C. It utilizes a policy network (Actor) to select actions and a value network (Critic) to evaluate the state. It improves stability by using the "advantage" function to reduce the high variance often seen in policy gradient methods.
2. **DDPG (Deep Deterministic Policy Gradient):** An off-policy, model-free algorithm adapted for continuous action spaces. It learns a deterministic policy and utilizes a replay buffer to store past experiences, effectively breaking temporal correlations in the training data.
3. **PPO (Proximal Policy Optimization):** A state-of-the-art policy gradient method that addresses the issue of unstable learning rates. It introduces a "clipping" function in the objective to prevent the new policy from deviating too drastically from the old policy during a single update, ensuring stable convergence.

Results and Discussion

The primary success metric is the Sharpe Ratio, measuring performance compared to a risk-free asset:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \tag{2}$$

Baseline Strategy: The performance of our RL agents was benchmarked against a passive "Buy & Hold" strategy. This strategy assumes purchasing the S&P 500 index (or an ETF tracking it, like SPY) at the beginning of the testing period and holding it until the end. This represents the average market performance.
Starting with an initial portfolio balance of $10,000 at the beginning of 2025, the models achieved the following results over the 1-year testing period:

Table 1: Final Performance Metrics (Out-of-Sample, 2025)

Strategy	Final Net Worth ($)	Return (%)	Sharpe Ratio
Buy & Hold (Baseline)	11,855.65	+18.56%	1.00
A2C Agent	12,444.20	+24.44%	1.31
DDPG Agent	12,059.16	+20.59%	1.32
PPO Agent	**12,506.12**	**+25.06%**	**1.37**

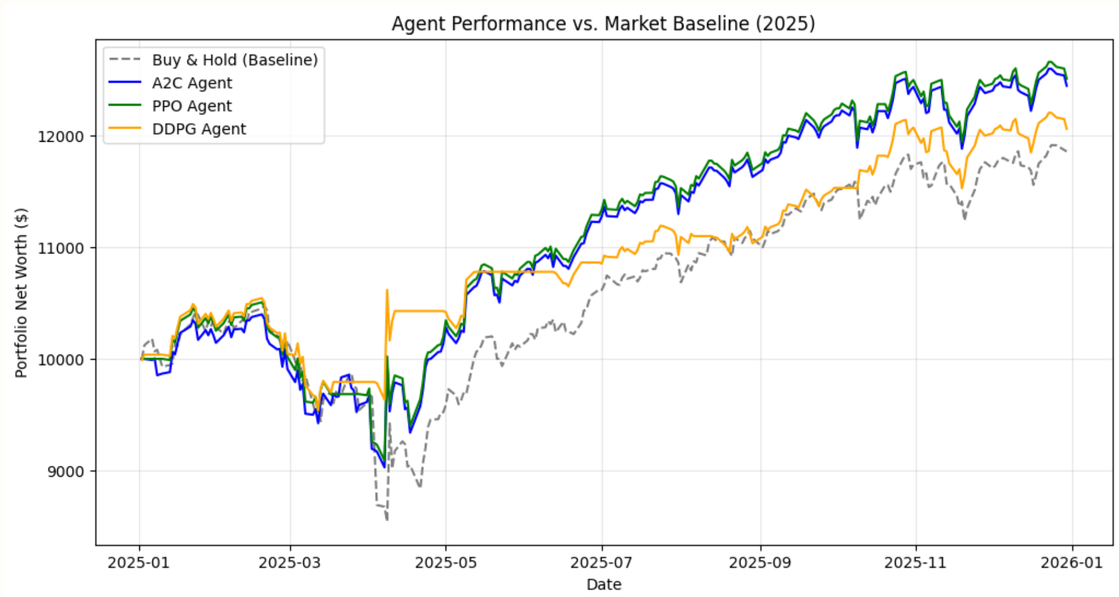

Figure 1: Portfolio Net Worth evolution of the DRL agents vs Baseline in 2025.

As shown in the table and figure above, all three DRL agents successfully outperformed the baseline strategy both in terms of absolute profit and risk-adjusted returns (Sharpe Ratio).
The **PPO agent** demonstrated the highest robustness and profitability. Thanks to its conservative policy updates (clipping), it effectively navigated market volatility and avoided large drawdowns compared to the pure Buy & Hold strategy.

Summary and Conclusions

This project demonstrates the viability of DRL for autonomous stock trading. Formulated as an MDP with a Logarithmic Alpha reward, our agents consistently outperformed the baseline, with PPO emerging as the most robust architecture. Future work includes expanding the action space to short selling, applying ensemble methods, and enabling intraday trading to allow multiple decisions per day.